Week 6 - Experimental Best Practices Checklist

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1. Selected Paper and Reproducibility Materials

The paper selected for this experimental best practices assessment was *"Supervised Machine Learning Algorithms for Predicting Student Dropout and Academic Success: A Comparative Study"* by Villar and de Andrade (2024), published in Discover Artificial Intelligence. It was selected because it is closely related to the doctoral research topic and evaluates several machine learning methods for predicting dropout, continued enrolment, and graduation in higher education.

The authors provide a public GitHub repository containing the Python notebook, documentation, statistical test data, and an image of the paired t-test results. The student dataset was downloaded separately from the UCI Machine Learning Repository under the title *"Predict Students' Dropout and Academic Success"* because the model dataset was not included directly in the GitHub repository.

The downloaded dataset contains 4,424 student records and is available under a CC BY 4.0 license. To align the current UCI version with the 35-column structure used in the original notebook, two additional variables were removed and the file reading parameters were adjusted. Some library installations and imports also required modification before the notebook could run in Google Colab. These materials formed the basis for assessing the study's experimental design, statistical rigor, data leakage risks, and computational reproducibility.

2. Reproducibility Scorecard

Checklist item	What to look for	Y	P	N
Seeds reported, many runs?	Mean ± SD, not one number	☐	☒	☐
Splits described?	Sizes, method, grouped or time-based if needed	☐	☒	☐
Leakage checked?	Preprocessing inside folds; no future features	☐	☐	☒
Tuning separated?	Validation set or nested CV	☐	☒	☐
Statistical test named?	Appropriate, paired when comparing models	☒	☐	☐
Effect size and CI?	Not just a p-value	☐	☐	☒
External validation?	New site, new period or new population	☐	☐	☒
Compute documented?	Hardware, runtime, versions	☐	☐	☒
Code and data available?	A stranger could re-run it	☐	☒	☐

3. Justifications

a) Seeds reported, many runs? — Partial

A random state of 42 is reported for the resampling procedures and most classifiers. However, the data split and optimization process are not fully seeded, and the results are not reported across multiple independent runs using measures such as the mean and standard deviation.

b) Splits described? — Partial

The notebook uses an 80% training and 20% test split. However, the split does not specify a random state or stratification, and the exact observations assigned to each subset cannot be reproduced consistently.

c) Leakage checked? — No

SMOTE and ADASYN are applied to the complete dataset before the training and test split. Consequently, synthetic observations are generated using information from records that may later appear in the test set, which can produce overly optimistic performance estimates.

d) Tuning separated? — Partial

Optuna and 10-fold cross validation are used for hyperparameter optimization, followed by evaluation on a test subset. However, the test subset is derived from data that were resampled before splitting, so it cannot be considered fully independent from the tuning and training process.

e) Statistical test named? — Yes

The study identifies a paired t-test for comparing model performance and provides the values used in the analysis and an image of the statistical result in the GitHub repository.

f) Effect size and CI? — No

The study reports statistical significance but does not provide effect sizes or confidence intervals. Therefore, the magnitude and precision of the differences between the evaluated models cannot be adequately assessed.

g) External validation? — No

All models are developed and evaluated using the same public student dataset. No independent dataset from another institution, population, or academic period is used to assess model generalizability.

h) Compute documented? — No

The repository provides the Python notebook and its library imports, but it does not report the hardware, operating system, Python version, package versions, execution time, or a reproducible environment file.

i) Code and data available? — Partial

The code and supporting statistical files are publicly available through GitHub, while the student data are available separately through the UCI Machine Learning Repository. However, the dataset is not included or directly linked in the repository, and the current UCI version requires adjustments to match the structure expected by the notebook.

4. Conclusion

The study obtained one Yes, four Partial, and four No assessments. The public code and clear identification of the statistical test support partial reproducibility. However, data leakage, incomplete control of randomness, limited computational documentation, and the absence of external validation reduce confidence in the reported performance. Future experiments should split the original data before applying resampling, perform resampling only within the training folds, fix all random seeds, report results from repeated runs, include effect sizes and confidence intervals, document the computational environment, and validate the final model using data from another institution or academic period.

5. References

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2021). *Predict students' dropout and academic success* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89

Villar, A. (n.d.). *SML-Comparative-Study* [Source code]. GitHub. Retrieved October 14, 2026, from https://github.com/alicevillar/SML-Comparative-Study

Villar, A., & de Andrade, C. R. V. (2024). Supervised machine learning algorithms for predicting student dropout and academic success: A comparative study. *Discover Artificial Intelligence, 4*, Article 2. https://doi.org/10.1007/s44163-023-00079-z